

CMPUT 461/561 Fall 2026

Lec 01 -- Brief overview of dialog systems

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Conversational agents and Chatbots

- Conversations and dialogues
- Agents vs Chatbots
- ELIZA and Neural chatbots

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Conversational agents / dialog systems

Recall that the goal of NLP is to bridge the gap between **human communication** and **computer understanding**.

- This is mostly done via dialog systems

A **conversation** is an interactive **communication** between two or more people.

A **dialogue** (sometimes spelled dialog in American English) is a written or spoken **conversational exchange** between (usually) two people, ...

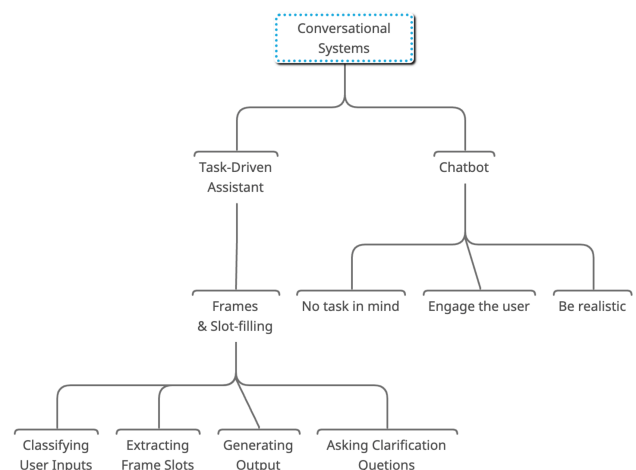
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Conversational agents

Conversations are the primary form of communication between humans.

– We are hardwired for conversation.

Human-Computer Interaction research has been pushing for Natural Language user interfaces for a long time.



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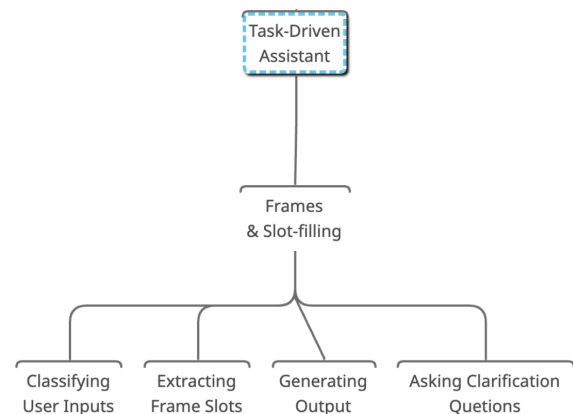
Task-driven assistant

Designed to accomplish specific tasks.

Engage in conversation with the user until all information needed for the task at hand is collected.

Frame = all information needed for a task.

Slot = piece of information in a frame.



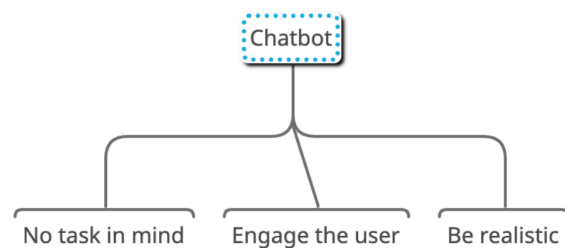
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Chatbot

Chatbots engage in conversation with the user for their entertainment/amusement.

- Are not built to accomplish any task.
- Can specialize in a topic/subject (ex: banking), and answer user questions.

Chatbots need to be **realistic** and produce high-quality language, to keep the users interested and engaged.



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Conversation Analysis

- Dialogs
- Conversation structure
- Dialogue acts
- Keeping track of the conversation

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Dialog

Sequence of alternating speech acts between **two agents**.

- The system knows it has to take turns with the human.

C₁: ...I need to travel in May.
A₂: And, what day in May did you want to travel?
C₃: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A₄: And you're flying into what city?
C₅: Seattle.
A₆: And what time would you like to leave Pittsburgh?
C₇: Uh hmm I don't think there's many options for non-stop.
A₈: Right. There's three non-stops today.
C₉: What are they?
A₁₀: The first one departs PGH at 10:00am arrives Seattle at 12:05 their time. The second flight departs PGH at 5:55pm, arrives Seattle at 8pm. And the last flight departs PGH at 8:15pm arrives Seattle at 10:28pm.
C₁₁: OK I'll take the 5ish flight on the night before on the 11th.
A₁₂: On the 11th? OK. Departing at 5:55pm arrives Seattle at 8pm, U.S. Air flight 115.
C₁₃: OK.
A₁₄: And you said returning on May 15th?
C₁₅: Uh, yeah, at the end of the day.
A₁₆: OK. There's #two non-stops ...#
C₁₇: #Act... actually #, what day of the week is the 15th?
A₁₈: It's a Friday.
C₁₉: Uh hmm. I would consider staying there an extra day til Sunday.
A₂₀: OK... OK. On Sunday I have ...

Figure 25.1 Part of a phone conversation between a human travel agent (A) and human client (C). The passages framed by # in A₁₆ and C₁₇ indicate overlaps in speech.

Speech Acts

speech acts

A key insight into conversation—due originally to the philosopher [Wittgenstein \(1953\)](#) but worked out more fully by [Austin \(1962\)](#)—is that each utterance in a dialogue is a kind of **action** being performed by the speaker. These actions are commonly called **speech acts** or **dialogue acts**: here's one taxonomy consisting of 4 major classes ([Bach and Harnish, 1979](#)):

Constatives:	committing the speaker to something's being the case (<i>answering, claiming, confirming, denying, disagreeing, stating</i>)
Directives:	attempts by the speaker to get the addressee to do something (<i>advising, asking, forbidding, inviting, ordering, requesting</i>)
Commissives:	committing the speaker to some future course of action (<i>promising, planning, vowing, betting, opposing</i>)
Acknowledgments:	express the speaker's attitude regarding the hearer with respect to some social action (<i>apologizing, greeting, thanking, accepting an acknowledgment</i>)

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Why does this matter?

The dialog system needs to maintain a representation of the "state" of the dialog; and choose the next speech act accordingly.

Task-oriented assistant

Keeps issuing **directives** to user until all information for the task is collected.

Chatbot

Chooses between different kinds of speech acts, as long as the user remains engaged.

constative →	C ₁ : ...I need to travel in May.
directive →	A ₂ : And, what day in May did you want to travel?
constative →	C ₃ : OK uh I need to be there for a meeting that's from the 12th to the 15th.
directive →	A ₄ : And you're flying into what city?
constative →	C ₅ : Seattle.

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Grounding

common
ground
grounding

A dialogue is not just a series of independent speech acts, but rather a collective act performed by the speaker and the hearer. Like all collective acts, it's important for the participants to establish what they both agree on, called the **common ground** (Stalnaker, 1978). Speakers do this by **grounding** each other's utterances. Grounding means acknowledging that the hearer has understood the speaker (Clark, 1996). (People need grounding for non-linguistic actions as well; the reason an elevator button lights up when it's pressed is to acknowledge that the elevator has indeed been called, essentially grounding your action of pushing the button (Norman, 1988).)

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Why does this matter?

The dialog system needs to be sure that the user has the same understanding of the "state" of the dialog as represented internally by the system.

Both assistants and chatbots need to generate language to confirm they understand the user and analyze what the user states to ensure grounding.

grounding

A₆: And what time would you like to leave Pittsburgh?
C₇: Uh hmm I don't think there's many options for non-stop.
A₈: Right. There's three non-stops today.
C₉: What are they?
A₁₀: The first one departs PGH at 10:00am arrives Seattle at 12:05 their time.
The second flight departs PGH at 5:55pm, arrives Seattle at 8pm. And the
last flight departs PGH at 8:15pm arrives Seattle at 10:28pm.
C₁₁: OK I'll take the 5ish flight on the night before on the 11th.
A₁₂: On the 11th? OK. Departing at 5:55pm arrives Seattle at 8pm, U.S. Air
flight 115.
C₁₃: OK.

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Dialogs (generally, conversations) are not always linear

side sequence
subdialogue

However, dialogue acts aren't always followed immediately by their second pair part. The two parts can be separated by a **side sequence** (Jefferson 1972) or **sub-dialogue**. For example utterances C₁₇ to A₂₀ constitute a **correction subdialogue** (Litman 1985, Litman and Allen 1987, Chu-Carroll and Carberry 1998):

C₁₇: #Act... actually#, what day of the week is the 15th?

A₁₈: It's a Friday.

C₁₉: Uh hmm. I would consider staying there an extra day til Sunday.

A₂₀: OK... OK. On Sunday I have ...

The dialog system needs to detect the start (and end) of sub-dialogs after each utterance.

The "state" of the sub-dialog may be nested within the state of the parent dialog.

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Task-driven Assistants

- Frames and slots
- Dialog structure
- Intent determination

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Frames and slots

In classical AI and knowledge representation, a **frame** is a data structure used to represent stereotypical situations or concepts.

- Think of it as `struct` in C or as a class in Java/Python.

Each **slot** in a frame corresponds to an attribute (or field) of the data structure.

Often, each slot is associated with a **question** for the user

Example: airline ticketing system

Frame: airline ticketing system

Slots:

FROM		What city are you leaving from?
TO		Where are you going to?
DEPART-DATE		When do you want to leave?
DEPART-TIME		At what time do you want to leave?
AIRLINE		Which airline do you prefer?
ARRIVE-TIME		At what time do you need to arrive?
CONFIRMED?		N/A

Some slots in the frame may not be associated with a question to the user. Ex: **CONFIRMED?** could be updated by the agent after receiving confirmation from the airline.

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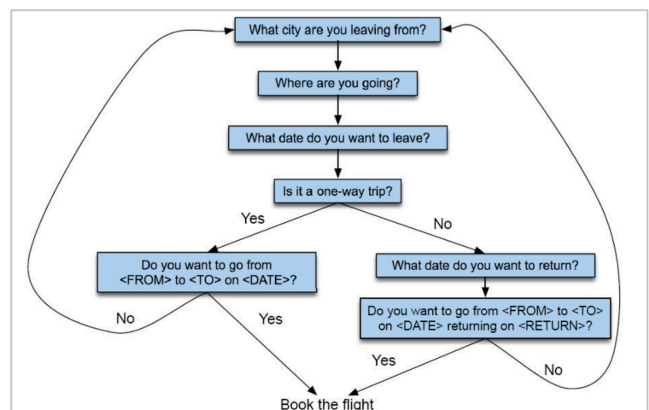
Single Initiative dialog systems

Early dialog systems had a **fixed** order of asking questions to fill out the slots.

Systems that control the conversation are called **single initiative**.

Most human interactions, initiative often shifts back and forth between the participants.

Single initiative is OK for very simple tasks only.



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Mixed initiative dialog systems

Mixed initiative systems allow the user to provide information in any order they want.

They still need to keep track of the state of the conversation.

Users are often much more satisfied with these systems.

Each user utterance must be analyzed independently.

```
GUS: Hello. My name is GUS. I can help you plan a simple trip by air. Where do you want to go ?
Client: I want to go to San Diego on May 28
GUS: What time do you want to leave?
Client: I must be in San Diego before 10 am
GUS: Would you like P.S.A. flight 102 that arrives at 9.15 am ?
Client: What is the next flight ?
GUS: Air California flight 310 that leaves at 8.30am and arrives at 10.00am
Client: I'll take the first one
GUS: In what name should I make a reservation, please?
Client: Dan Bobrow
GUS: I have confirmed the following flight: P.S.A. flight 102 on Wednesday May 28 from San Jose to San Diego that leaves at 7.30 am and arrives at 9.15 am.
GUS: What date do you want to return on ?
Client: On Friday in the evening.
GUS: Would you like the flight that leaves at 7.45 pm ?
Client: That's fine.
GUS: I have confirmed the following flight: P.S.A. flight 307 on Friday May 30 from San Diego to San Jose that leaves at 7.45 pm and arrives at 9.30 pm Thank you for calling. Goodbye
```

Adapted from Jurafsky & Martin

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Information extraction from sentences

Information Extraction is the task of extracting intended information from sentences (or larger pieces of text).

- Besides slot filling, it is also used in building knowledge graphs.

Example:

Show me morning flights from Boston to San Francisco on Tuesday
a system might want to build a representation like:

DOMAIN:	AIR-TRAVEL	INTENT:	SHOW-FLIGHTS
ORIGIN-CITY:	Boston	DEST-CITY:	San Francisco
ORIGIN-DATE:	Tuesday	ORIGIN-TIME:	morning

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Understanding utterances in dialogs

dialogue acts **Dialogue acts** are a generalization of speech acts that also represent grounding. The set of acts can be general, or can be designed for particular dialogue tasks.

Tag	Sys	User	Description
HELLO($a = x, b = y, \dots$)	✓	✓	Open a dialogue and give info $a = x, b = y, \dots$
INFORM($a = x, b = y, \dots$)	✓	✓	Give info $a = x, b = y, \dots$
REQUEST($a, b = x, \dots$)	✓	✓	Request value for a given $b = x, \dots$
REQALTS($a = x, \dots$)	✗	✓	Request alternative with $a = x, \dots$
CONFIRM($a = x, b = y, \dots$)	✓	✓	Explicitly confirm $a = x, b = y, \dots$
CONFREQ($a = x, \dots, d$)	✓	✗	Implicitly confirm $a = x, \dots$ and request value of d
SELECT($a = x, a = y$)	✓	✗	Implicitly confirm $a = x, \dots$ and request value of d
AFFIRM($a = x, b = y, \dots$)	✓	✓	Affirm and give further info $a = x, b = y, \dots$
NEGATE($a = x$)	✗	✓	Negate and give corrected value $a = x$
DENY($a = x$)	✗	✓	Deny that $a = x$
BYE()	✓	✓	Close a dialogue

Figure 15.6 Dialogue acts used by the HIS restaurant recommendation system of [Young et al. \(2010\)](#). The Sys and User columns indicate which acts are valid as system outputs and user inputs, respectively.

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Utterance	Dialogue act
U: Hi, I am looking for somewhere to eat.	hello(task = find, type=restaurant)
S: You are looking for a restaurant. What type of food do you like?	confreq(type = restaurant, food)
U: I'd like an Italian near the museum.	inform(food = Italian, near=museum)
S: Roma is a nice Italian restaurant near the museum.	inform(name = "Roma", type = restaurant, food = Italian, near = museum)
U: Is it reasonably priced?	confirm(pricerange = moderate)
S: Yes, Roma is in the moderate price range.	affirm(name = "Roma", pricerange = moderate)
U: What is the phone number?	request(phone)
S: The number of Roma is 385456.	inform(name = "Roma", phone = "385456")
U: Ok, thank you goodbye.	bye()

Figure 15.7 A dialogue from the HIS System of [Young et al. \(2010\)](#) using the dialogue acts in Fig. 15.6.

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"Complete" example

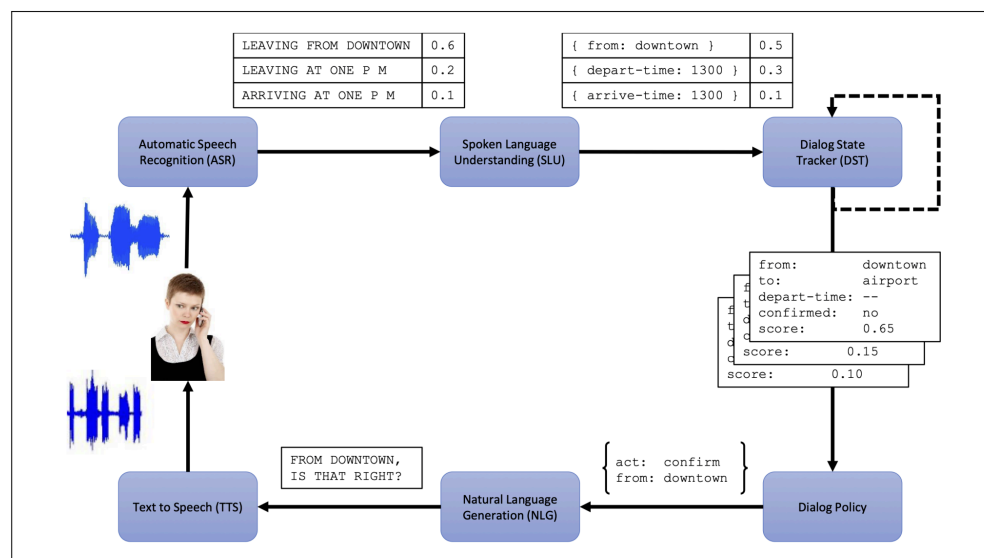


Figure K.1 Architecture of a dialogue-state system for task-oriented dialogue from Williams et al. (2016).

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Dialog management (state tracker)

Part of the "logic" of the assistant (i.e., not really an NLP task).

Designed to suit the goals of the assistant.

- Frames and rules to check if the assistant has enough information to act.
- Remembers previous utterances.

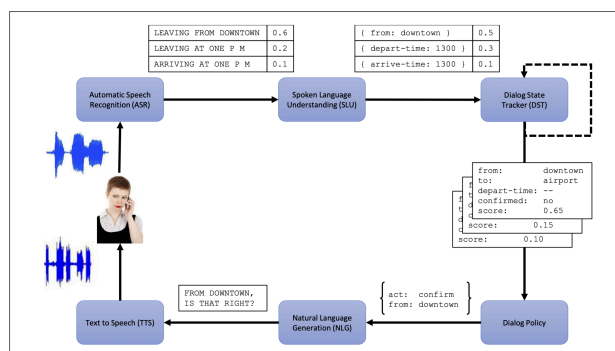


Figure K.1 Architecture of a dialogue-state system for task-oriented dialogue from Williams et al. (2016).

```
while (true):
    intent = detect_intent()
    while (not enough information):
        ask_relevant_question()
    if intent=quit:
        exit()
```

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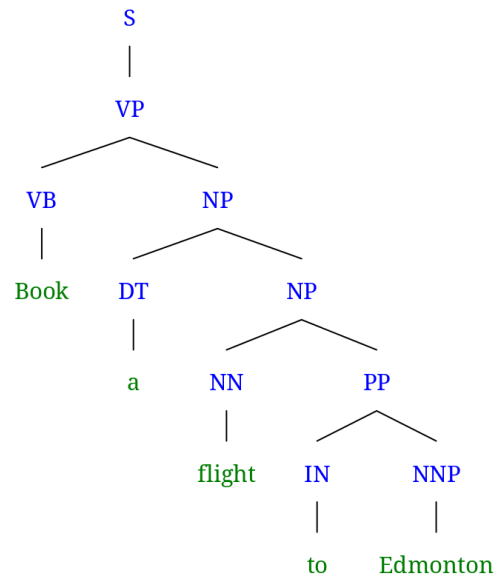
Pre-LLM NLP with parsing

Classical NLP methods relied on parsing to understand the sentences.

Intent classification can be done by looking at the verbs.

Slot filling is also possible!

- Navigate the three to find (proper) nouns affected by the verb.
- We will see a different (and more suitable) form of parsing for this.



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RECAP : pre-LLM and LLM-based intent classification

pre-LMM using Machine Learning

Collect data: thousands of sentences that express the intent, and thousands that do not.

Pre-process: tokenize them and represent them as vectors using word embeddings.

Intent classification: train a model to predict the intent from the vectors.

pre-LMM rule based

Design an algorithm.

With LLM

Zero shot: Just ask an LLM to do it for you.

Prompt engineering: use a more "conductive" prompt.

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Early phone assistants (ex: SIRI)

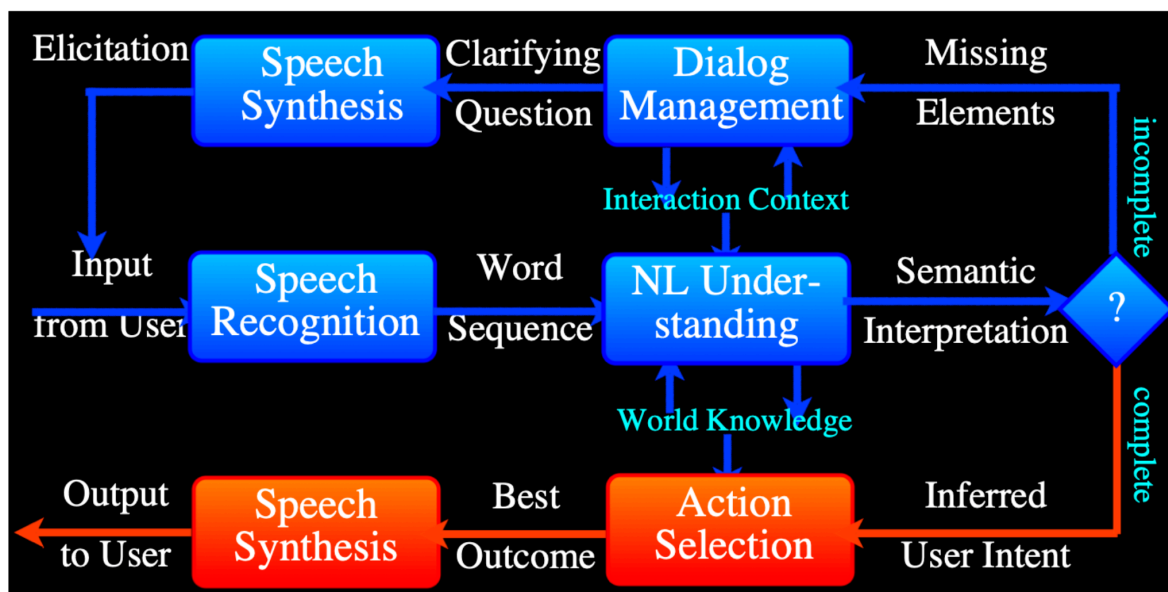
Designed as traditional assistants that could handle many different tasks.

- Make a phone call.
- Play music.
- Activate IoT devices.
- Tell jokes.
- Book appointments; ...

Are now being replaced by LLM-based chatbots.

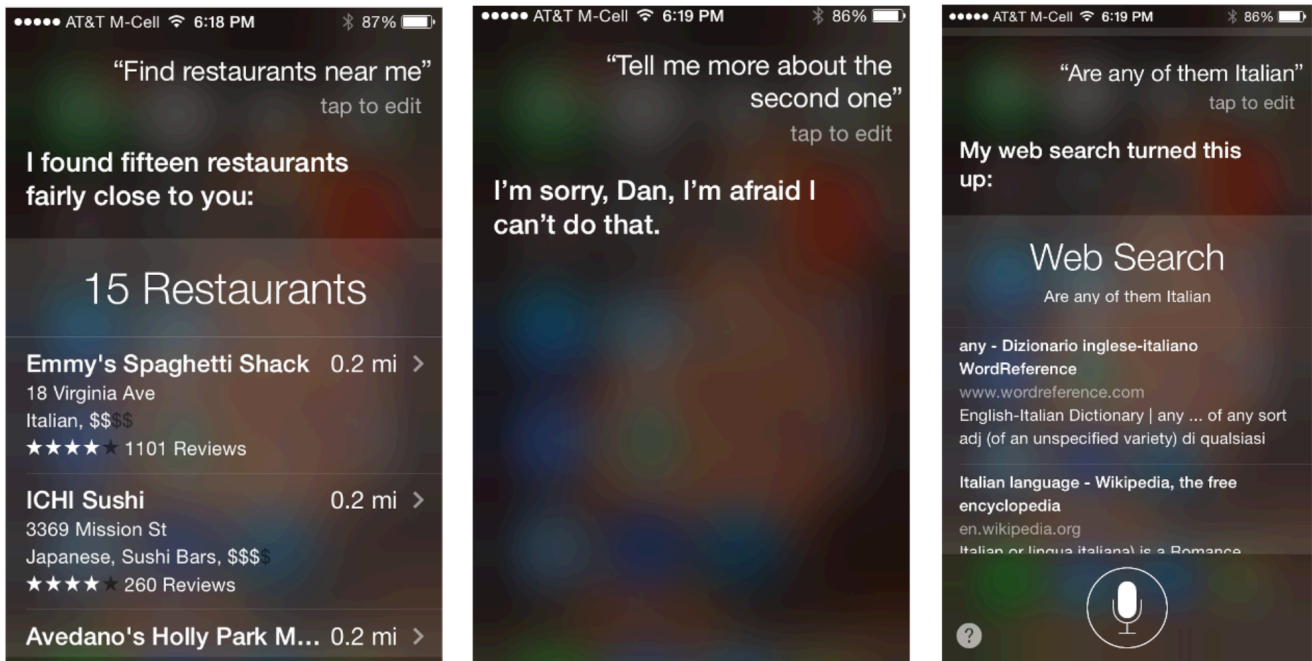
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Typical architecture

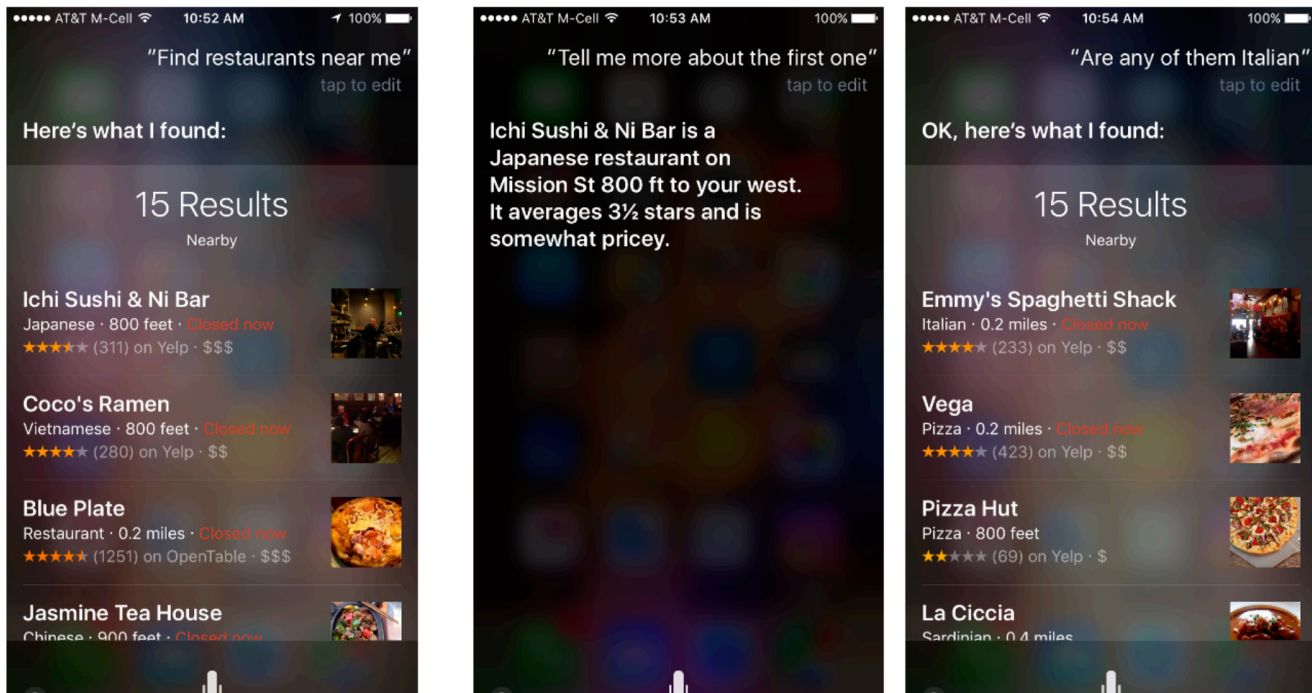


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Example: SIRI as of 2014



Example: SIRI as of 2017



Take-away from examples

Building conversational agents is hard!

SIRI 2014

✓ Search the web for what the user wanted (restaurants) and based on context (location).

✗ Poor dialog management.

- Ex: failed to differentiate between the "first" and "second" restaurants.

✗ Understand complex user requests.

- (Ex: find if the restaurants were Italian).

SIRI 2017

Addressed both deficiencies in the earlier version of SIRI.

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Chatbots

- ELIZA
- Typical LLM chatbot

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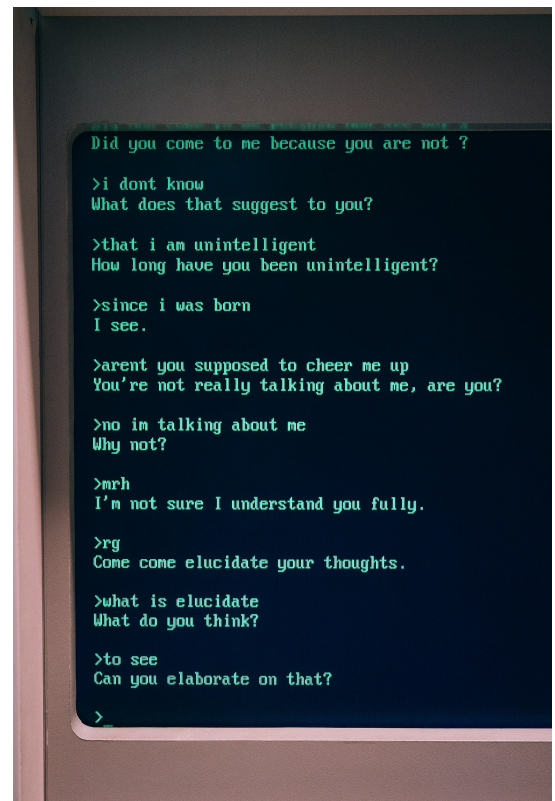
ELIZA

Early NLP system developed at MIT by Joseph Weizenbaum in the early 1960s to explore communication between humans and machines.

Simulated conversations via **pattern matching and substitution**

- Gave users an illusion of understanding.
- Had real understanding of the conversation.

<https://web.njit.edu/~ronkowitz/eliza.html>



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ELIZA as a Rogerian Psychologist

ELIZA had many "scripts". The most famous is the one where it simulates a psychologist.

- Tendency to turn back to the human (patient) to talk more about the subject.

Human: I went on a long boat ride.
ELIZA: Tell me about boats.

The human engages, not knowing that ELIZA has no idea of what a boat is.

Rogerian psychotherapy is a humanistic approach psychotherapy centered on: unconditional positive regard (acceptance), congruence (genuineness), and **empathic understanding**.

https://en.wikipedia.org/wiki/Person-centered_therapy

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Among Weizenbaum's clever tricks are the linking of each ELIZA pattern/rule to a keyword. Consider the following user sentence:

I know everybody laughed at **me**

Because it has the word "I", this sentence could match the following rule whose keyword is *I*:

I (.*) -> You say you \1

producing:

YOU SAY YOU KNOW EVERYBODY LAUGHED AT **YOU**

Weizenbaum points out, however, that a more powerful response would rely on the keyword "everybody", since someone using universals like *everybody* or *always* is probably thinking about a specific person or situation. So the ELIZA algorithm prefers to respond using patterns associated more specific keywords like *everybody*:

WHO IN PARTICULAR ARE YOU THINKING OF?

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Pattern matching with regular expressions

ELIZA relied on the state-of-the-art sentence processing tool of its time: regular expressions.

- Match utterances with certain *keywords* (I, you, me, ...) of interest to psychologists.
- Generate responses via **substitutions**.

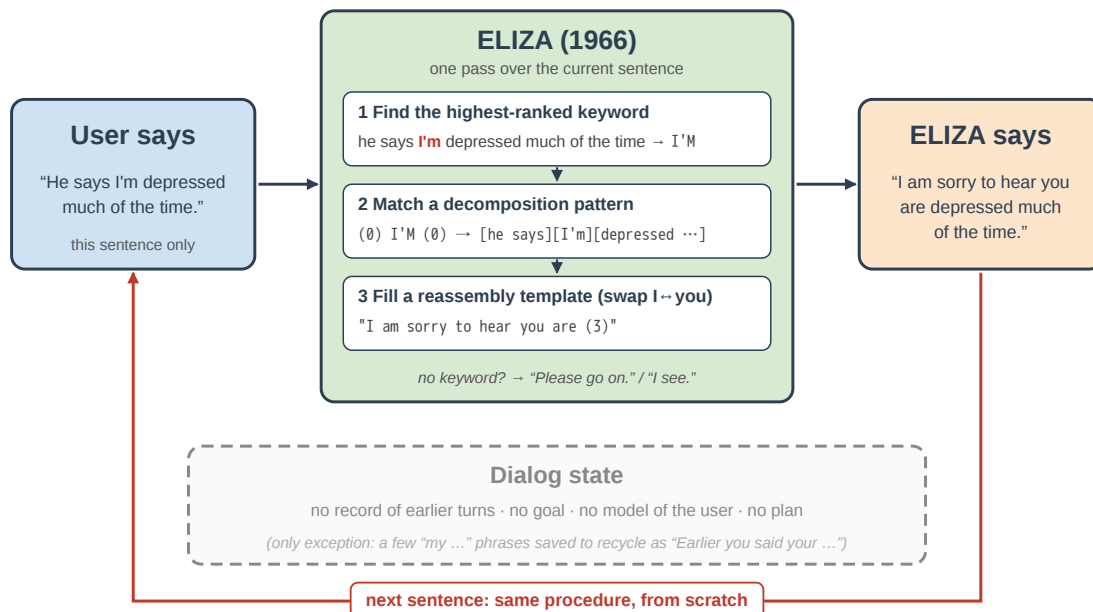
Each response is based only on the previous utterance by the user.

(.*) YOU (.*) ME -> WHAT MAKES YOU THINK I \2 YOU

to transform a user sentence like "You hate me" into a system response like
WHAT MAKES YOU THINK I HATE YOU

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Dialog management in ELIZA (there is NONE!)



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Ethical issues arising from ELIZA

People became emotionally involved with the program.

- Weizenbaum told that his secretary would require him to leave the room before she could engage with ELIZA.
- When he suggested archiving conversations for later analysis, people immediately reported privacy concerns.

Anthropomorphism is the attribution of human form, character, or attributes to non-human entities. It is considered to be an **innate tendency of human psychology**.

<https://en.wikipedia.org/wiki/Anthropomorphism>

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Neural chatbots

Similarities with ELIZA

No predefined intended task.

Aimed at being **convincing** and keeping the user engaged.

No explicit modelling of the conversation.

Ethical issues arising from their use.

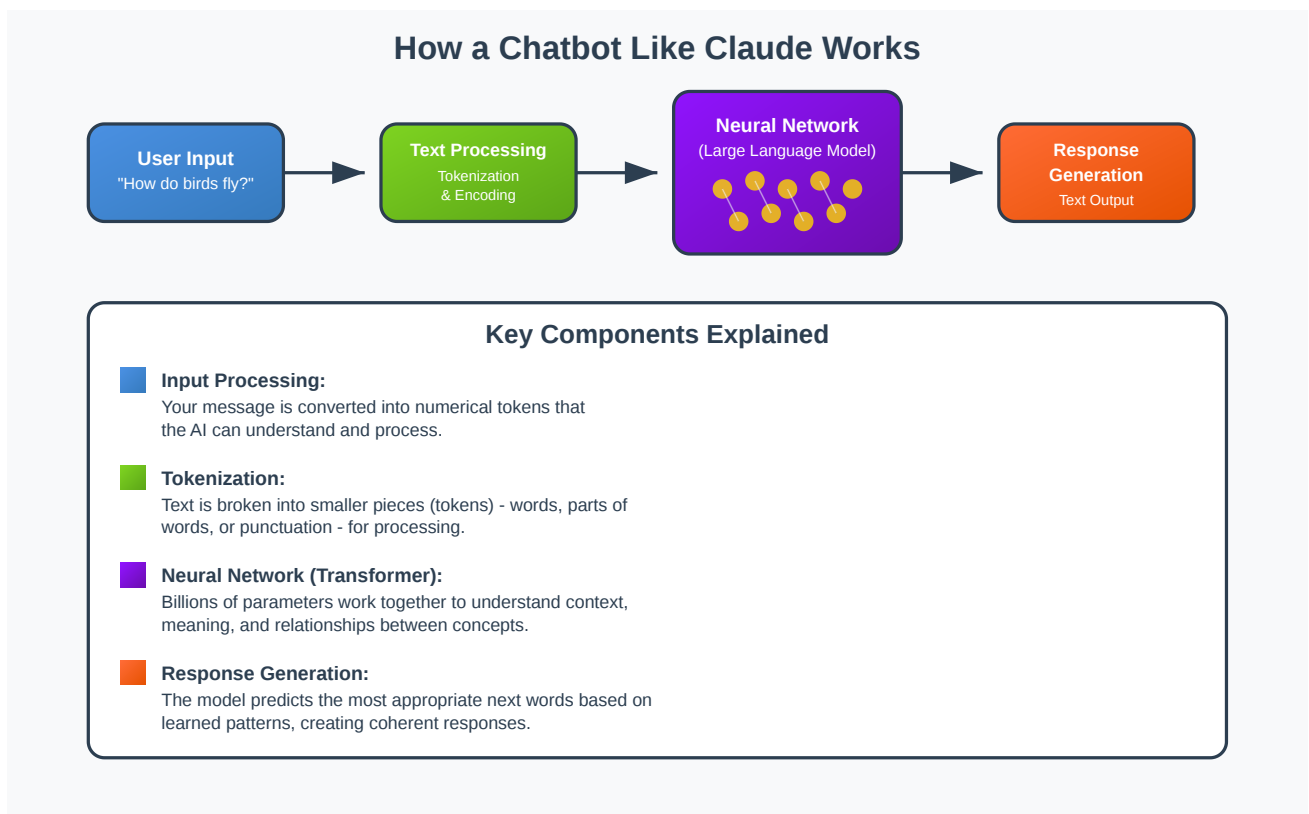
Differences to ELIZA

Response generated by a neural network trained on a lot of data (and, potentially fine-tuned for a domain) instead of regular expressions.

Much larger context, including several of the past interactions (as opposed to a single sentence).

Do have a dialog management module.

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Language model

Any mechanism used to predict the "next" word in a sentence.

Unidirectional

"I ate the ____"

w	$P(w)$
pizza	0.25
apple	0.18
sandwich	0.12
homework	0.005
blue	≈ 0

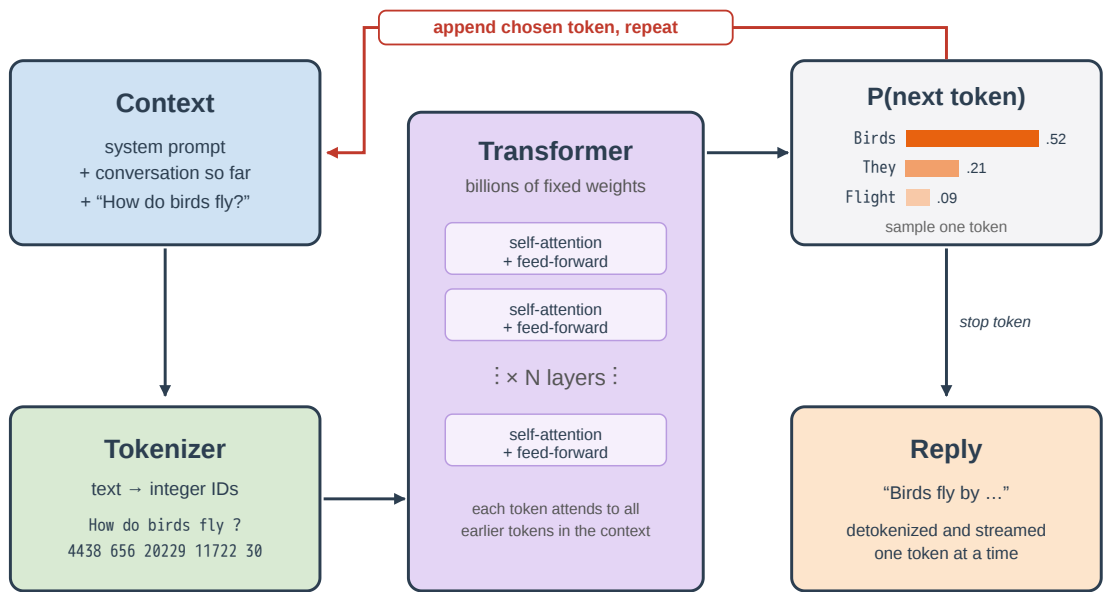
Bidirectional

"I took my ____ to the park"

w	$P(w)$
dog	0.35
ball	0.28
friend	0.20
table	0.001
cat	≈ 0

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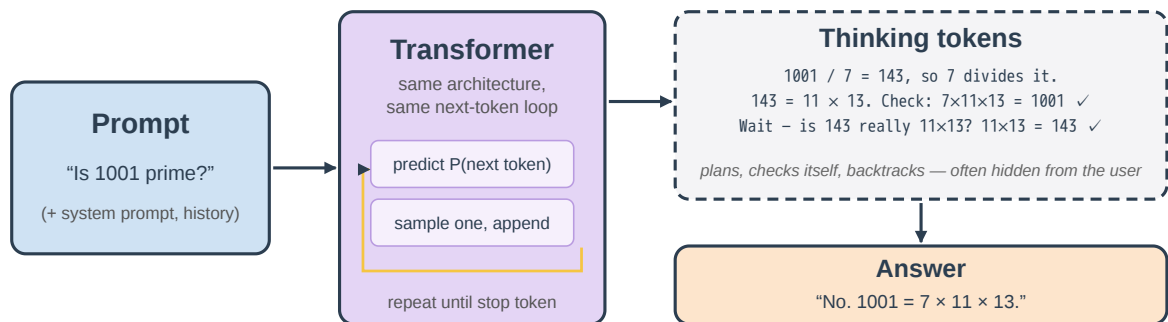
LLMs generate one word at a time



Weights were learned once by predicting the next token over trillions of tokens, then fine-tuned on dialogues. Nothing is looked up at chat time.

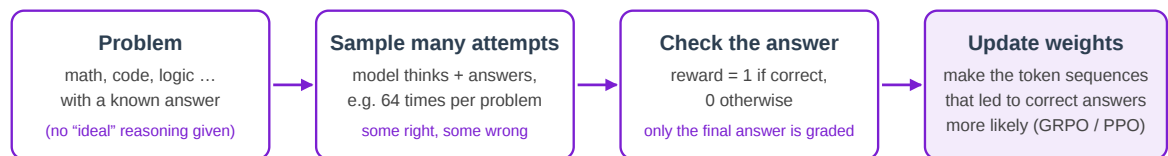
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"Reasoning" LLM



One token stream: the thinking tokens are in the context when the answer is written.

How it learned to think: reinforcement learning on problems with checkable answers



Over millions of problems the model discovers on its own that longer, self-checking thinking earns more reward.
The length of that thinking is where extra compute at inference time goes ("thinking budget").

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For later

These will be covered in later classes:

Classical methods

- Regular expressions
- PoS tagging and parsing (constituency, dependency, semantic).

LLMs

- Training of neural chatbots.
- Risks of using chatbots (bias, hallucination, pleasing humans).

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Relationship to other modules

1. Lexical and syntactical ambiguity
2. PoS tagging
3. Parsing, semantic parsing
4. Classification (rule-based, machine-learning-based)
5. Language generation
6. Large language models

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What you should know or be able to do

- Explain the similarities and differences between chatbots and assistants
- Describe the architecture of a task-oriented assistant
- Explain what are frames and slots
- Explain different methods of determining user intent
- Explain how assistants and chatbots keep track of the "conversation" with a human
- Explain risks of training bots and assistants with texts found on the web
- Explain which parts of an assistant are easier or harder to do with an LLM

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